Solder Paste Inspection

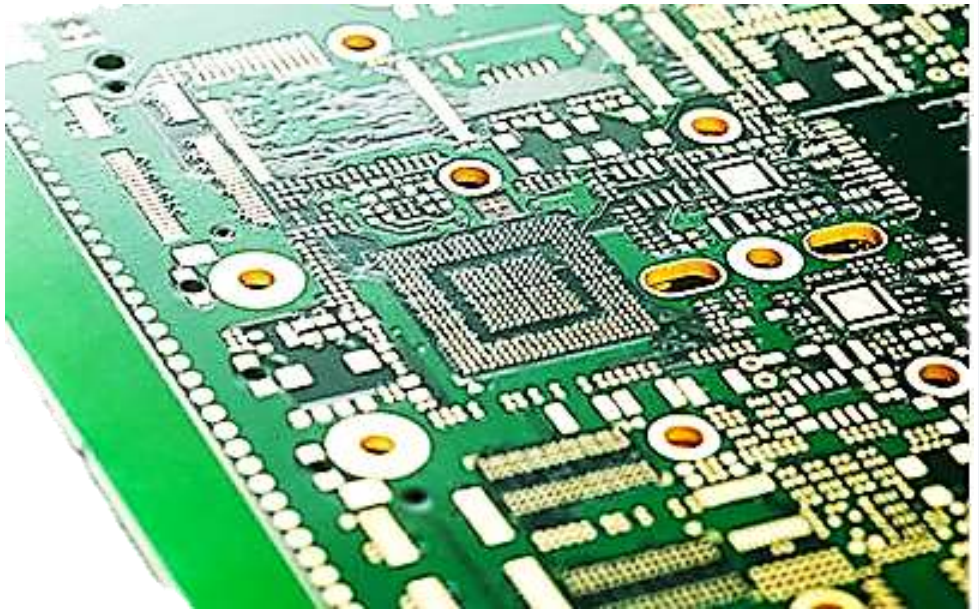
(S P I)

❑ Introduction

- In Solder Paste Technology (SMT) solder paste deposition is vital process in board assembly technology. This process is prone to defects and errors during manufacturing process. Defects can be found either in solder paste itself or in the solder printing process.

- With conventional methods ,fixing defects was very time consuming as defect identifications were very subjective, and decision taken resulting in material wastage and excessive downtime.

- Considering high-density mounting, it is necessary to decrease the pad size as much as possible and narrow the gap between the devices. As per the growing demand, technology change witnessed in Solder paste, printing options, component placement and reflow profiling.



Package		Dimensions, length × width	
Metric	Imperial		
0201	008004	0.2 mm × 0.1 mm	0.0078 in × 0.0039 in
0402	01005	0.4 mm × 0.2 mm	0.0157 in × 0.0079 in
0603	0201	0.6 mm × 0.3 mm	0.024 in × 0.012 in
1005	0402	1.0 mm × 0.5 mm	0.039 in × 0.020 in

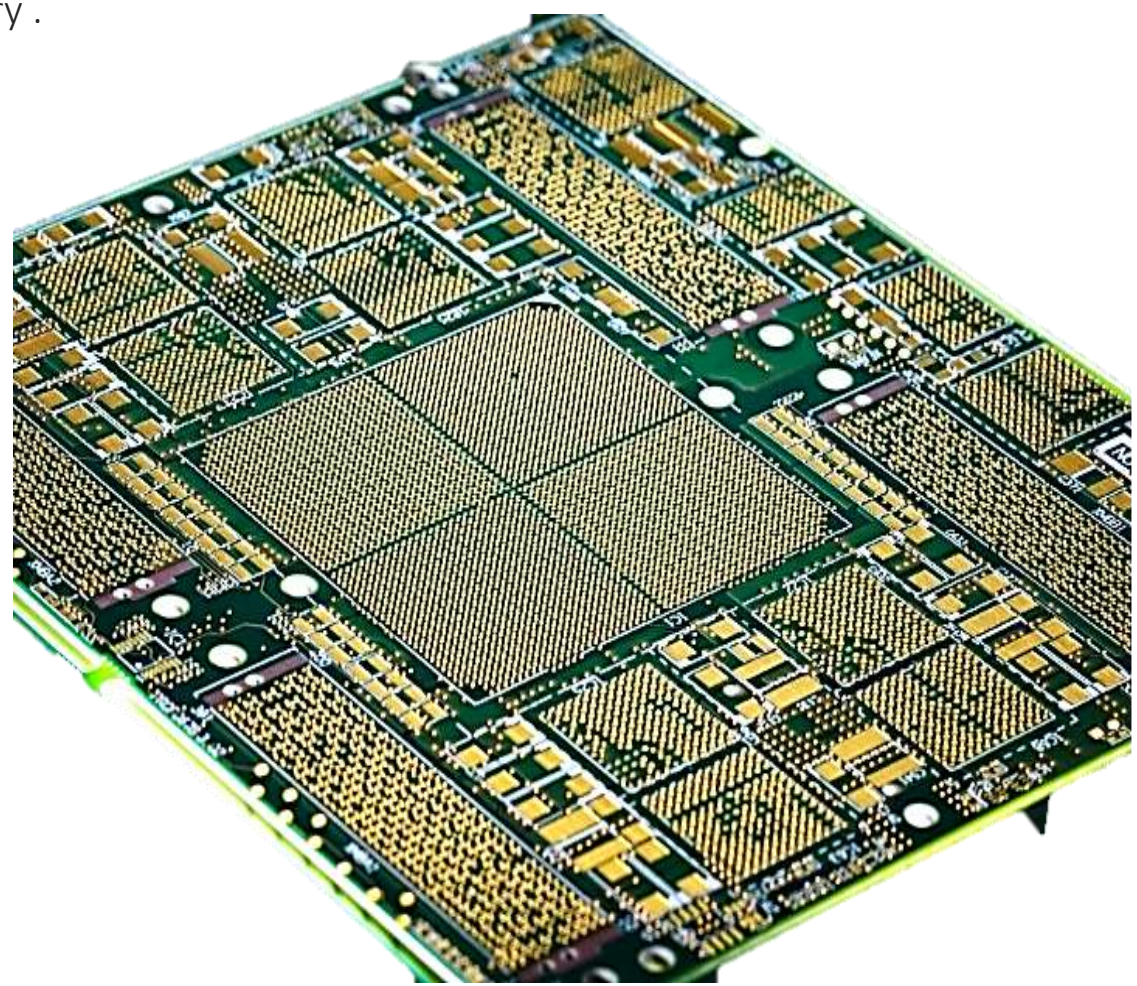
- Reduction in size of surface mount devices from 0805 -0603-0402- 0201 - 01005 and still further, and pitch of the components decrease with tighter tolerances, taking risk of print defect has become a big challenge . So, human intervene on 100% inspection on every Board on production line not possible

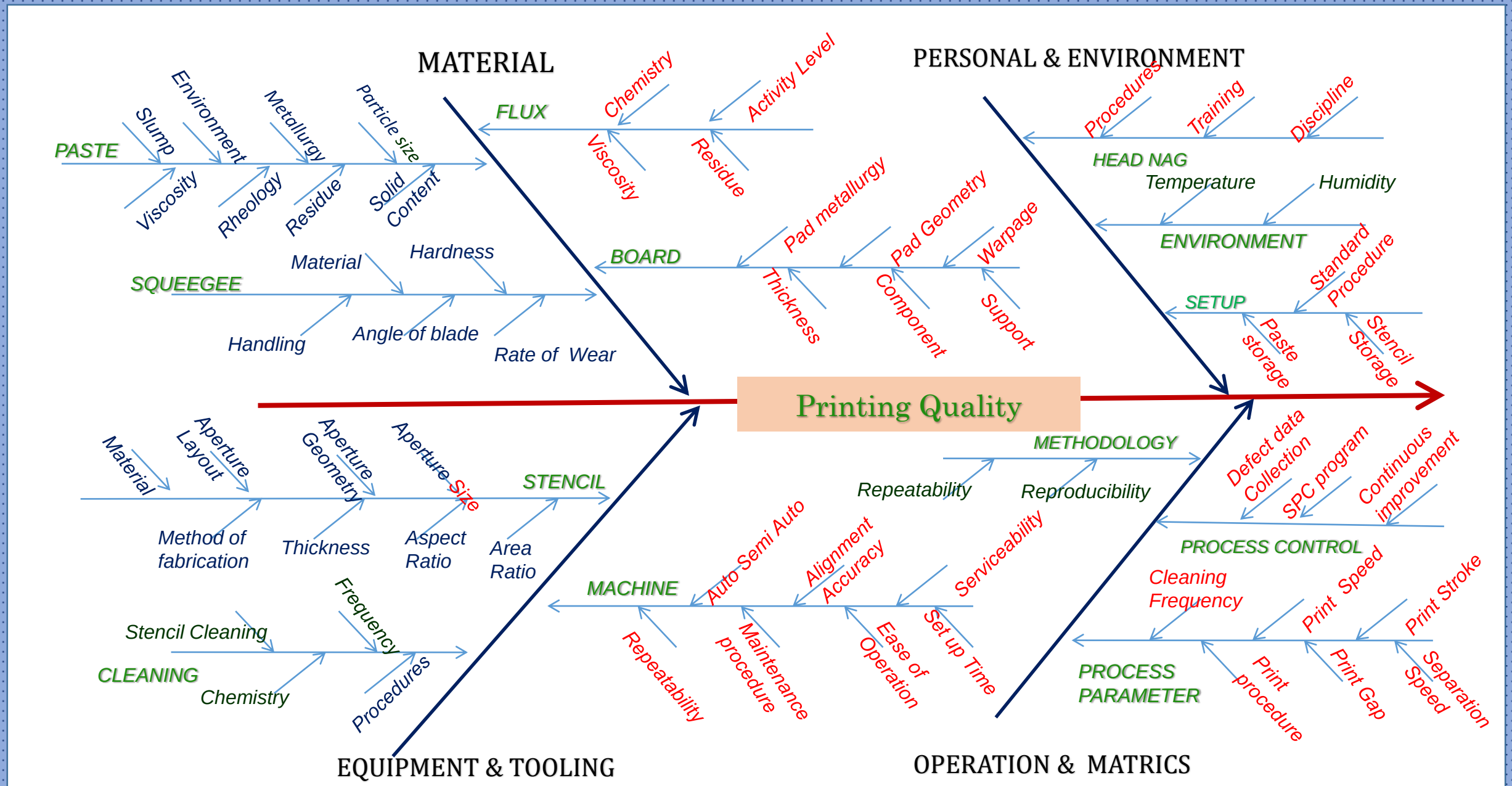
❑ Introduction cont..

- Increased use of Ball Grid Arrays(BGA), Fine pitch and Ultra fine pitch Quad pack packages (QFPs), Package on package (POPS), there is a dramatic increase in demand solder paste inspection after stencil printing. Demand for 100% inspection , online corrective actions become mandatory .

BGA	µBGA, CSP, PBGA, CBGA, CCGA, CLGA, LBGA, SBGA,T BGA &WLP
Ball Pitch	Fine pitch – 0.5mm, 0.75mm, 0.8mm Standard Pitch – 1mm, 1.27mm, 1.5mm
Ball Día	0.4mm to 0.75mm
Ball count	2x2 (4 balls) To 50 x 50 (2500 balls)
Size	0.6mm To 15mm

QFP	TQFP LQFP QFN
Lead Pitch	Fine pitch – 0.3mm, 0.4mm, 0.65mm – 1mm i.e., 11.8 mils TO 40 mils
Lead count	32 leads To 304 leads




MATERIAL
PERSONAL & ENVIRONMENT
EQUIPMENT & TOOLING
OPERATION & MATRICS
Printing Quality
PASTE
Slump
Environment
Metallurgy
Particle size
Viscosity
Rheology
Residue
Solid Content
FLUX
Chemistry
Activity Level
Viscosity
Residue
SQUEEGEE
Material
Hardness
Handling
Angle of blade
Rate of Wear
BOARD
Pad metallurgy
Pad Geometry
Warpage
Thickness
Component
Support
Procedures
Training
Discipline
HEAD NAG
Temperature
Humidity
ENVIRONMENT
SETUP
Standard Procedure
Paste storage
Stencil Storage
STENCIL
Material
Aperture Layout
Aperture Geometry
Aperture Size
Method of fabrication
Thickness
Aspect Ratio
Area Ratio
CLEANING
Stencil Cleaning
Frequency
Chemistry
Procedures
MACHINE
Auto-Semi Auto
Alignment Accuracy
Serviceability
Repeatability
Maintenance procedure
Ease of Operation
Set up Time
METHODOLOGY
Repeatability
Reproducibility
PROCESS CONTROL
Defect data Collection
SPC program
Continuous improvement
PROCESS PARAMETER
Cleaning Frequency
Print Speed
Print Stroke
Print procedure
Print Gap
Separation Speed

❑ Purpose of Solder Paste Measurement

- The purpose of a measurement machine is to provide data accurately represent real product dimensions as this is responsible for most assembly defects.
- Industry research shows that solder paste volume is linked to joint quality and reliability: too much or too little translates into unreliable joints - an unacceptable outcome.
- Measurements need for production line to closely monitor solder paste alignment and volume during printing process as a part of quality management system .
- Provide key data about printing process i.e., Defect location on each PCB, Top 10 defects , real timer monitoring (line performance), 3D defect image, Cp/Cpk and various detailed Pareto and Pie charts which allows us to clarify Major key performance indicators.
- Exact measurement of the height of the solder paste in the printing process.
- Measuring solder paste volume can predict the quality and long-term reliability of solder joints.

❑ Measurement Fundamentals

❖Accurately measuring solder paste deposit heights and volume depends on several factors

- Image sensors resolution
- PCB surface undulations
- Height threshold settings
- PCB optical characteristics
- Post process algorithm that computes height and volume

❑ Measurement Variations

- Instrument variations
- Measurement process variations
- Operated related variations
- Variations among Test samples
- Environmental Fluctuations

❑ Accurate Measurement Technical Hitches

- PCB surface smoothness
  - •Local Warpage
  - •Undulation on Surface
  - - Results in uncertainty of the reference height of the PCB.
    - Small deposits more sensitive to warpage
- Paste deposit Location
  - Metal pad is a mix of three intersecting surfaces Metal Pad , Solder Mask an FR4 Solder Material. This complicates reference plane for measurement. identifying and setting up a
- Solder paste deposit – shape defining – cubic or Cylinder

S P I Machine Components

- SPI Machine
  - Head Section
    - X Y Z Axis
    - Probe Unit
      - PZT Amplifier
      - 3D LED L – R Projection
      - 2D LED
      - RGB Light
  - Conveyor Section
  - Power Unit
    - DC Supply
    - MCB
    - Contactors
    - Safety Relays
  - User Interface
  - Control Unit
    - GNET - M Controller — X Y Z Axis Control
    - GNET - D Controller — Conveyor Moment Control
    - GNET - R Controller — I/O Signal Control



❖ PZT Amplifier – Piezo transducer which moves the Grating unit . Machine takes 8 image of pad and calculates average values.
Left projection ON – Capture image 1 – Grating Movement – Left projection ON – Capture Image 2 - -- (4 images) – Right projection ON – Capture image 1 -----(4 Images) Total 8 Images captured .

3D LED left and right Projection – White LED used to calculate paste

2D LED to read fiducial

RGB Light – Used for fiducial reading + Barcode reading + PCB true colour image

Noise filter – Any Noise or surge killing for incoming AC voltage

GNET M controller – used to drive the axis servomotors of X, Y and Z with respect to their individual axis servo amplifiers

GNET D controller – (STMC board Older Version) Used to control DC stepper motors of 1. PCB loading (conveyor rotation) 2. width adjustment motors

GNET R controller – 1. Controls all I/O signals such as Sensors and Solenoids . 2. Controls servo signals 3. controls I/O signals like SMEMA, Barcode trigger etc.

Power supply – 12 V DC for camera power, 24V DC for sensors and solenoids and 5V DC for GNET controller Cards.

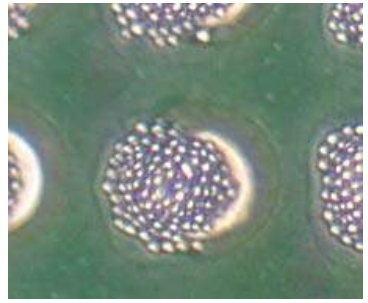
Why 3D Inspection?

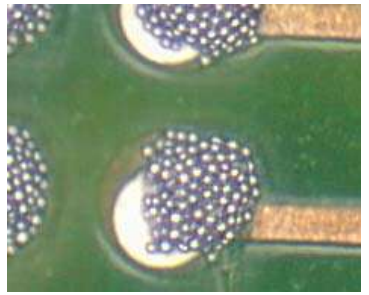
- 3D inspection on all boards and pads is required because the solder-joints reliability and strength are determined by the volume and shape of solder paste.

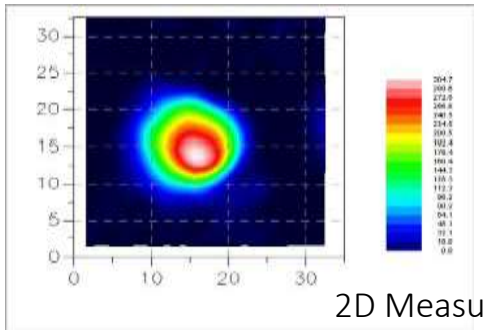
- Current market are familiar with fine pitch parts and 0603, Six Sigma Movements, which make 3D (solder paste) inspection necessary.

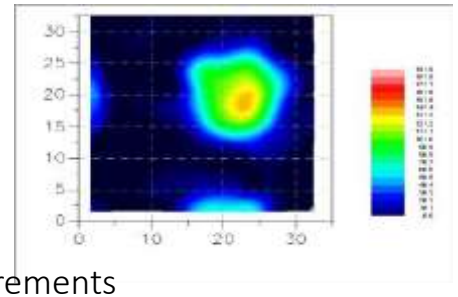
❖ Two deposits looks very similar In terms of area

❖ But they are far different values in volumes

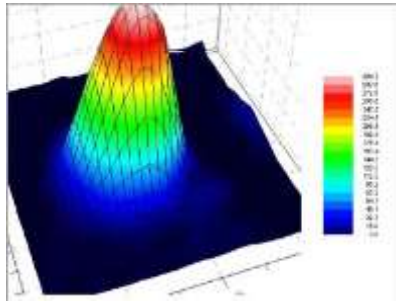


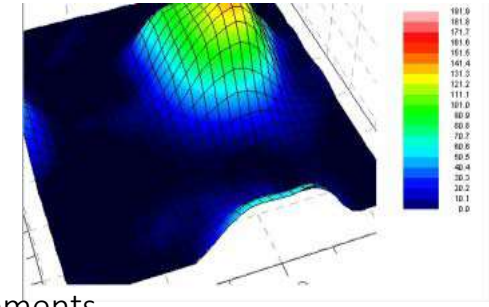






2D Measurements





3D Measurements

Area Vs Volume

❑ Understanding Working Principle of SPI -

❖ SPI utilizes angle cameras that take rapid 3D pictures to measure solder paste volume and alignment. The working principle of 2D and 3D both are same but 2D measures only XY offset whereas 3D measures offset as well as height. 3D captures the Height of solder paste (also known as “Z” for Z axis) which enables the equipment more accurately measure total volume of paste printed.

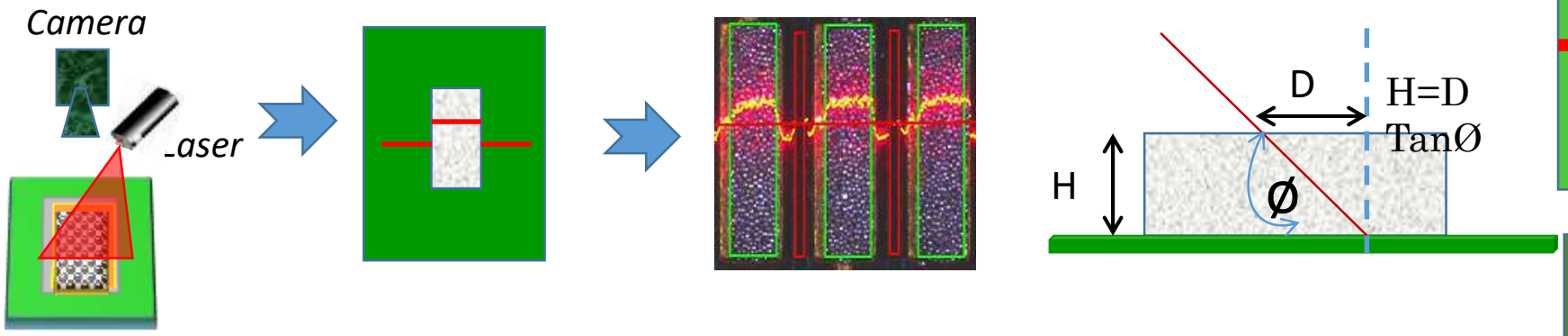
❖ The core technologies for solder paste Inspection systems.

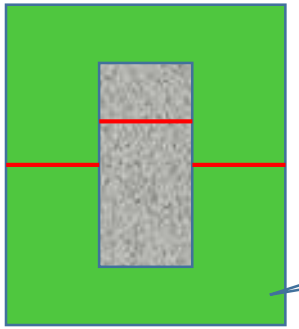
- laser-based
- phase shift or White light Moire´ interferometry.
  - Moiré based, stripes projection
  - More accurate and repeatable
  - No laser safety related issue
  - White light projection through power LEDs and dual projection makes the inspection indifferent to change of colors and materials.

Laser Based Technology -How does laser-based technology work?

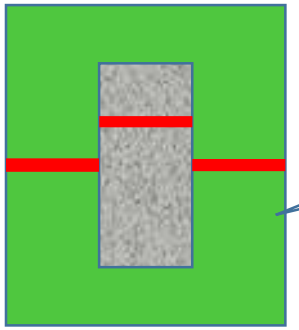
- Utilizing the core principles of laser triangulation, a laser stripe is projected at a specific angle to the imaging camera. As the laser moves across the surface, a string of data is automatically collected from the centroid of the laser stripe generating a differential height reading from the base surface to the overall surface of the solder paste deposit

- Triangulation displacement uses a laser beam to measure solder volume without contact. With the triangulation method, each pixel can obtain distance images consisting of height data. This distance image, which calculates the solder volume, is a 3D profile of the printed solder.

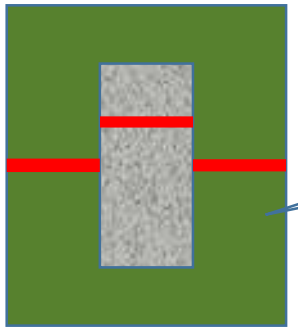




- *Low intensity, better focus, better measurement accuracy.*
- *Slower and more accurate*



High intensity , brighter line, faster and in accurate.



Higher intensity required for darker PCB. Slower and in accurate.

❖ Technology Concerns

- Not accurate, Not repeatable
- Specula problem is critical
- Weak against the optical noise Vibration noise is critical
- Laser Intensity can affect the result of the measurement result.

  A lot of fine tuning required by user.
- Laser makes a single scan, delivering a single data point per pixel –no possibility to identify, if data is good or bad!
- Safety precautions are needed due to the laser

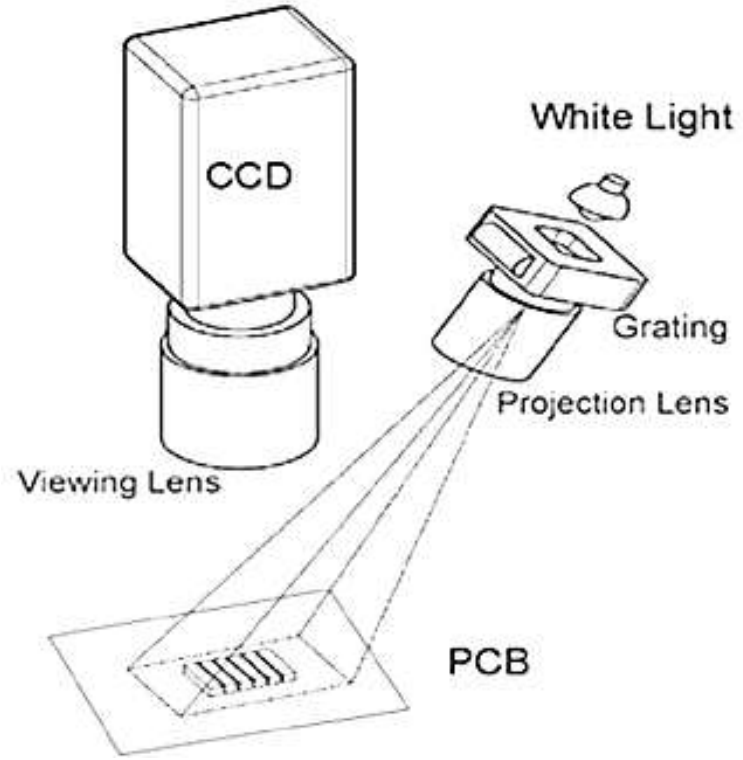
❑ Phage Shift Or Moiré Technology

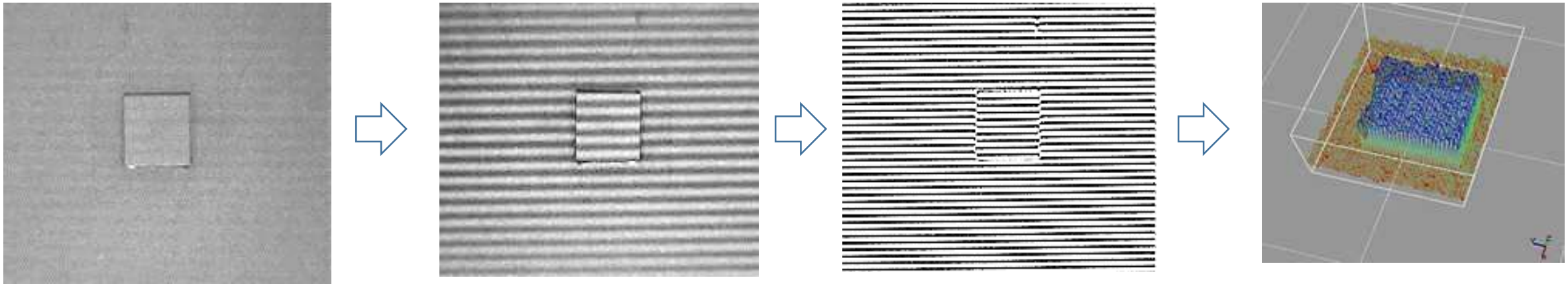
1. White light source projects a pattern of well-defined periodic fringes on to object (PCB)

2. 2Mpixel CCD Camera captures the projected image for four times in sequence , which includes height information of the object by pixel .

3. As those four frames of captured images are being processed , their height information significantly amplified by interference between four image frames and the reference.

4. It enables us to create 3D shape image of all solder paste deposits , measure volume and monitor shape deformity and so on…





(Courtesy KOHYOUNG)

Bottlenecks of the Optical Triangulation Principle

- There was two well-known problems in the field of three-dimensional shape measurement and inspection

1. Shadow effect - Pastes shaped as slope which are steeper than the projected pattern or light cannot be measured and can only be guessed.

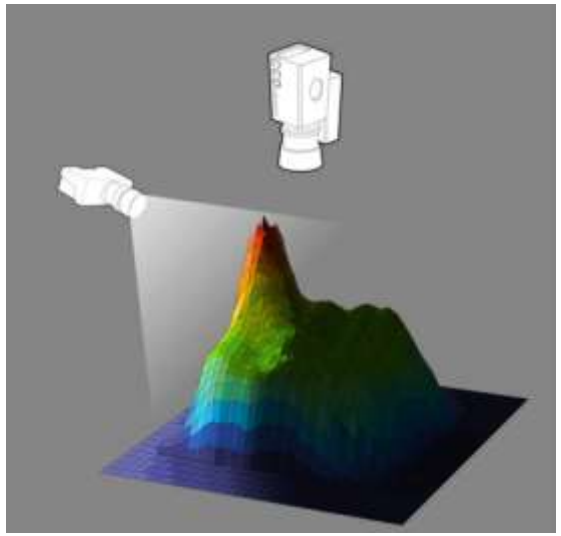
2. Specular area - Some pastes may give specular reflection of the projected pattern or light to te camera or sensor. As a result, we can expecht at most unreliable or noisy 3D measurement data for those area.

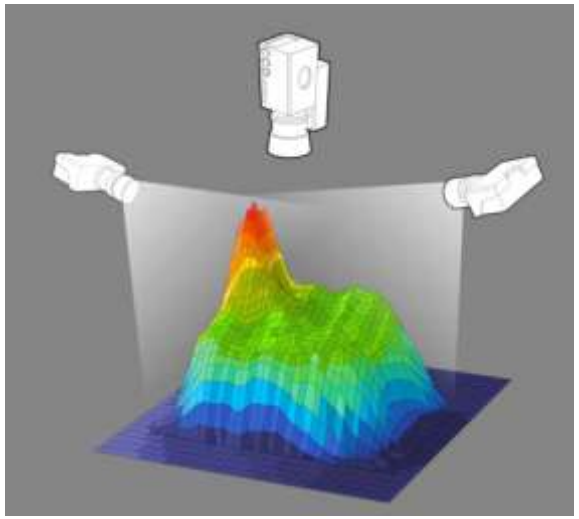
Solution for Bottle necks

1. Capture the Right set of the 3D image data consisted of 4 image frames : (R1, R2, R3, R4)

2. Capture the Left set of the 3D image consisted of 4 image frames : (L1, L2, L3, L4)

3. At each pixel, compute the “S/N Ratio Formula” to get PR and PL;

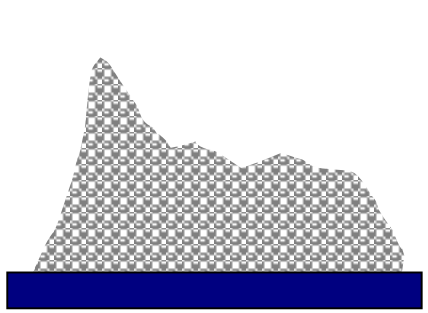


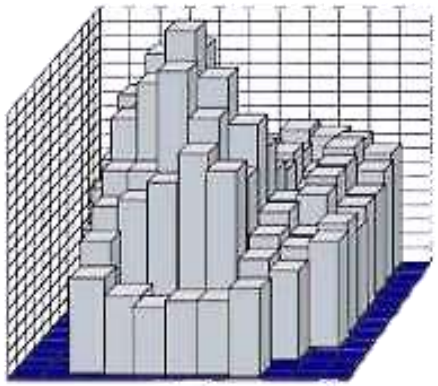


❑ Definition of Volume

- Using a true 3D profilometry method, Kohyoung can measure the authentic volume of solder pastes on PCB's, not rough approximation or simplification.

- Koh Young computes each pixel height of solder paste and sums up the unit volume of each pixel :



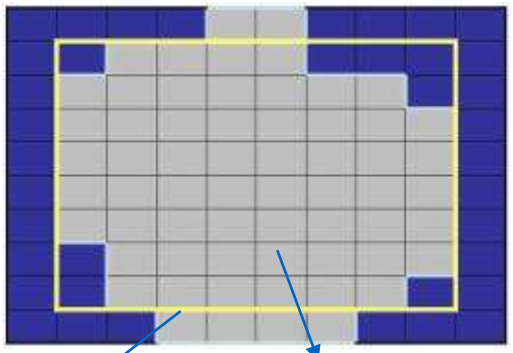


$$\text{Height} = \text{Average}\ (h_{(i,j)})_{(i,j)\in Solder\ Area}$$

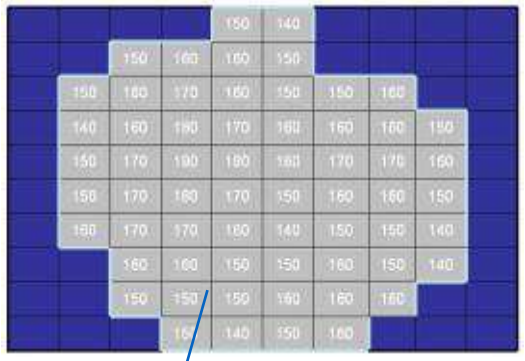
$$\text{Measured Volume} = \sum_{(i,j)\in SolderArea} h_{(i,j)} \times Area\ of\ one\ pixel$$

$$\%\ \text{Volume} = \frac{Measured\ Volume}{\underbrace{Stencil\ thickness \times Area\ of\ Pad}_{\text{Target Volume}}} \times 100$$

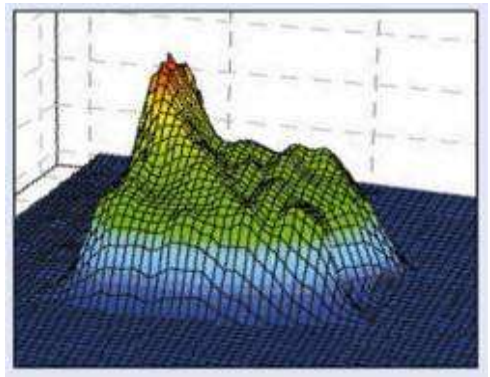
- X,Y resolution: 20 ㎛



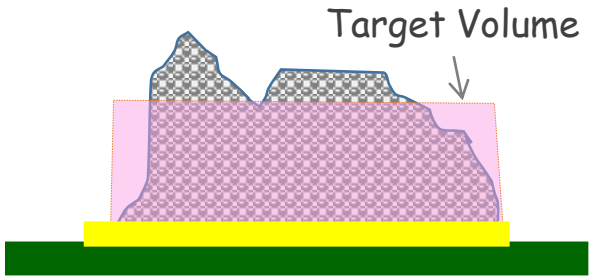
Desired area Measured area



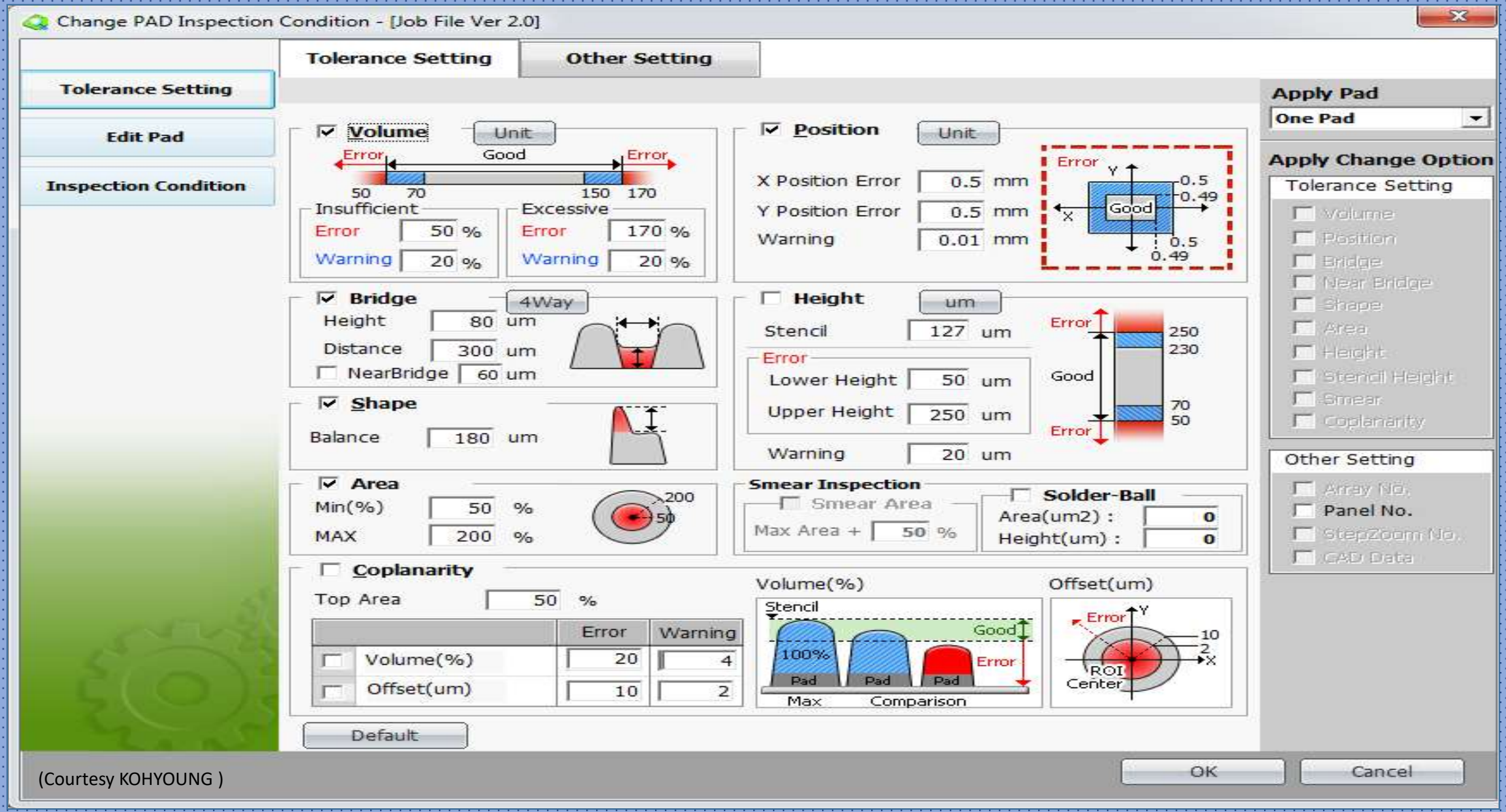
Measured height per pixel



Koh Young's 3D Viewer



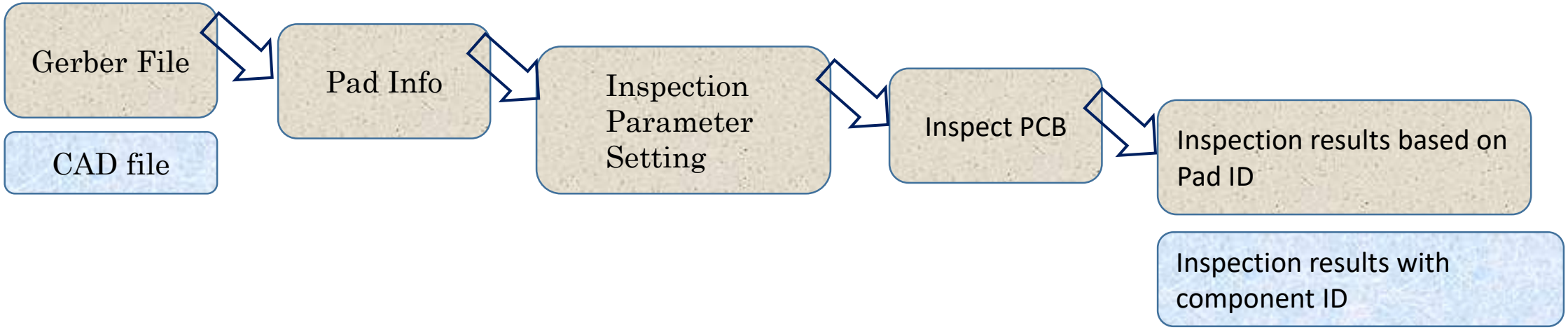
$$Area_{Stencil\ opening} = Area_{Solder\ Paste}$$



(Courtesy KOHYOUNG)

❑ Programming

- Making Program have an options either Online or offline
- Automated solder paste equipment requires mainly Gerber data
- Software will take care of Conversion, assignment of fiducials and inspection parameter set-up. Inspection results & statistics based on PAD numbering
- Optional use of CAD data, combined with stencil Gerber which provides data of Inspection results & statistics have component & pin ID.



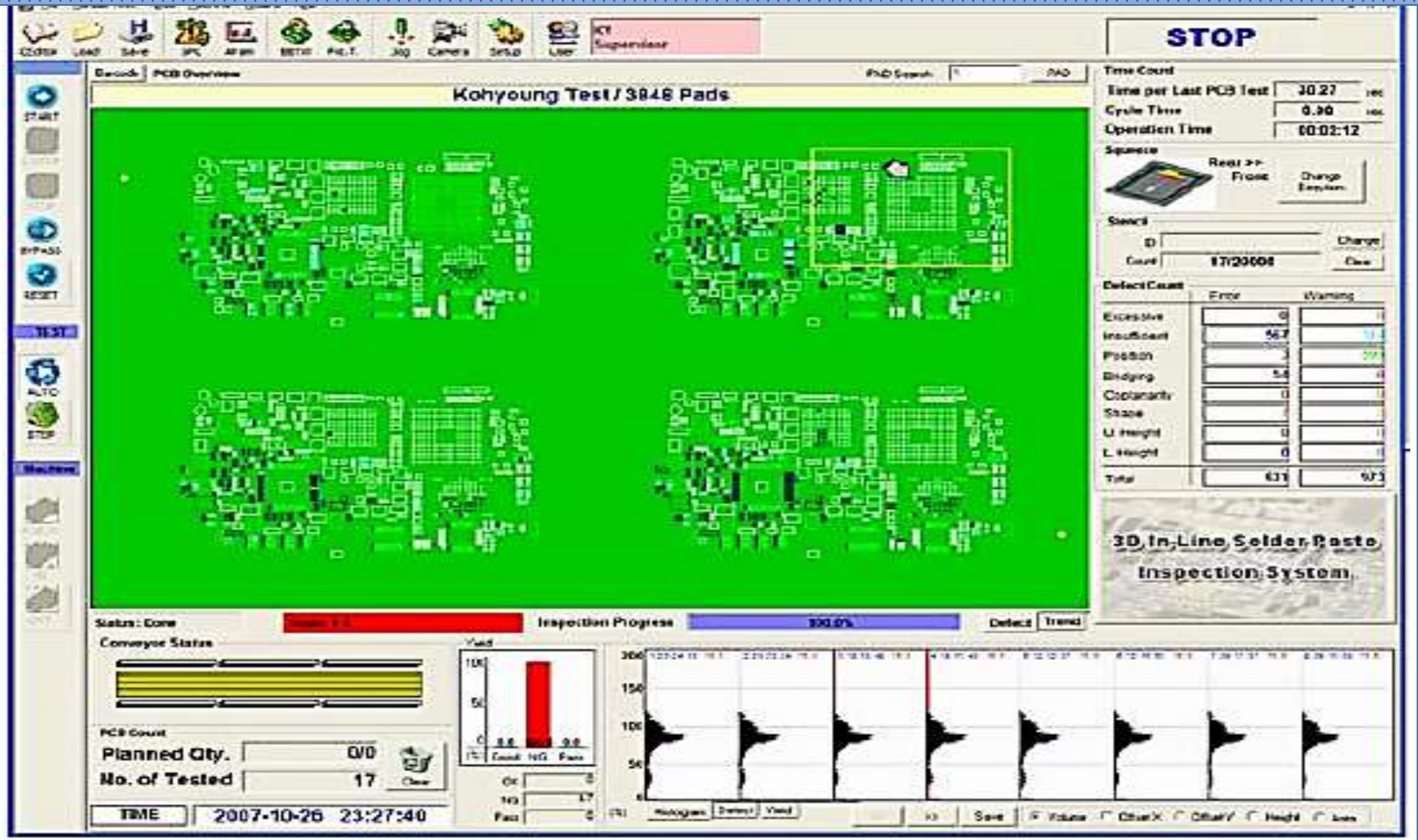
❖ Gage Repeatability and Reproducibility (GR&R)

An industry accepted practice to assess the repeatability of SPI platforms is a Gage R & R study which entails a single board with paste deposits placed in an oven at 105 to 125°C for approximately 1 hour to drive off the liquids and solidify the paste deposits. The board is then run through the SPI platform multiple times including rotations of 90, and 180 degrees

Real time Summary Chart - All In One page

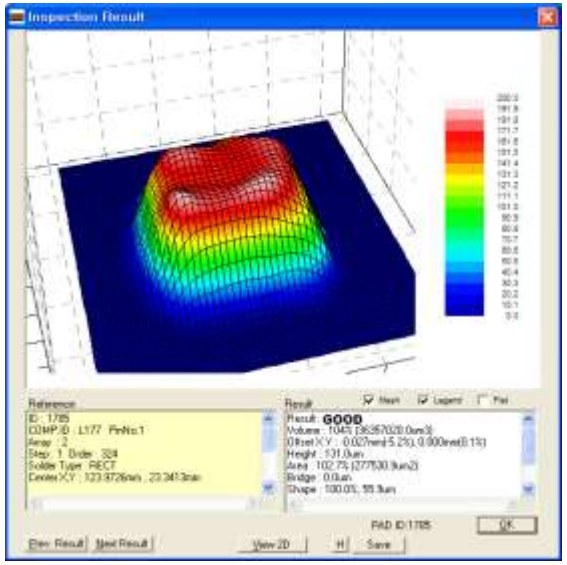
Example - Glance through this for understanding purpose



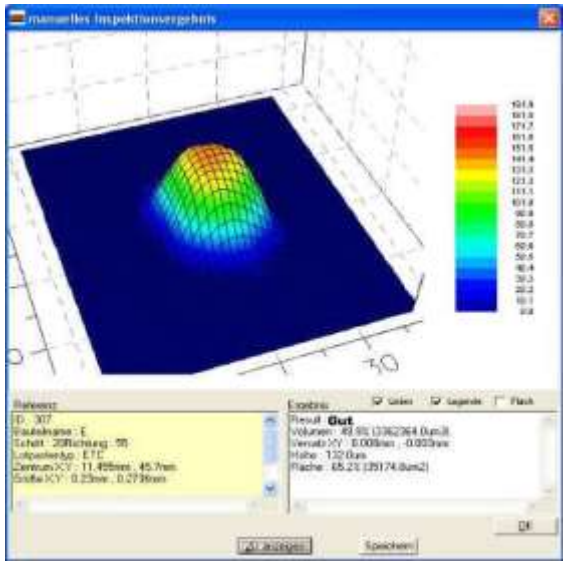

STOP
Kohyoung Test / 3848 Pads
Time Count
Time per Last PCB Test
Cycle Time
Operation Time
00:02:12
Squeeze
Rear >>
Front
Change Direction
Stencil
ID
Change
Clear
Defect Count
Error
Warning
Excessive
Insufficient
Position
Bridging
Coplanarity
Shape
U Height
L Height
Total
3D In-Line Solder Paste
Inspection System
Status : Done
Inspection Progress
Defect
Trend
Conveyor Status
PCB Count
Planned Qty.
0/0
No. of Tested
17
Clear
TIME
2007-10-26 23:27:40
Histogram
Trend
Yield
Save
Height

Defect Detection - Examples

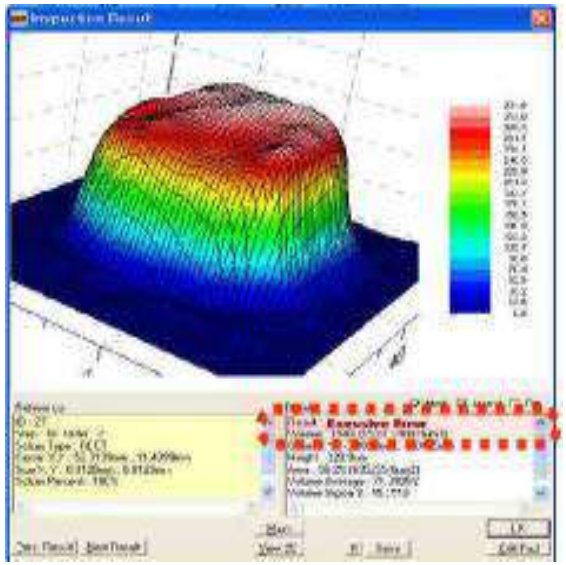
Normal



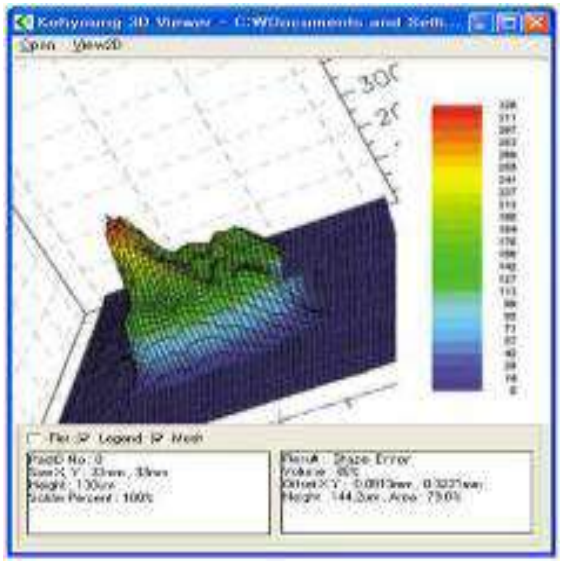
Insufficient

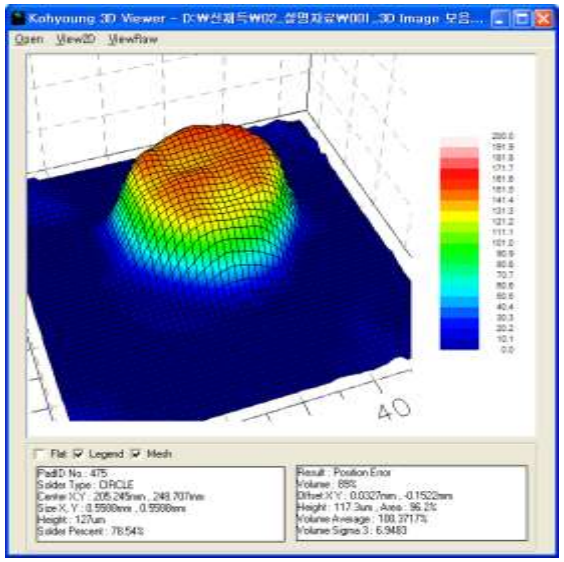


Excessive



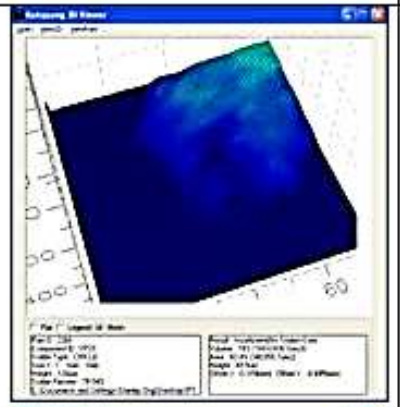
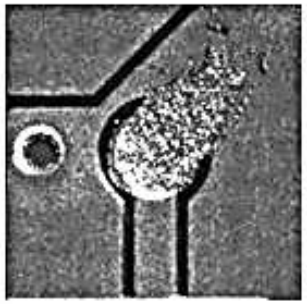
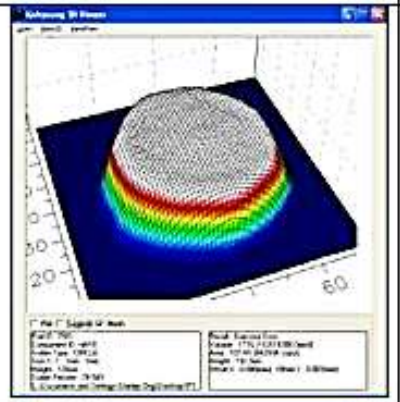
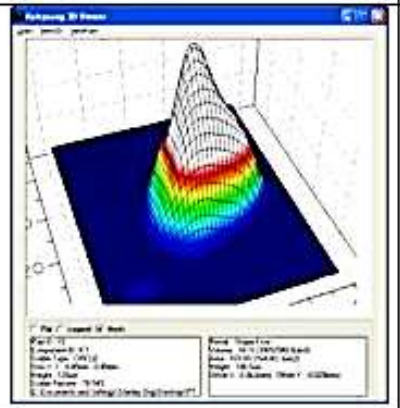
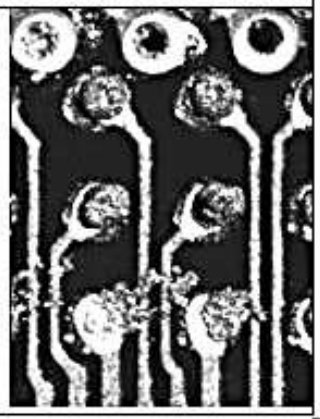
Shape Deformity

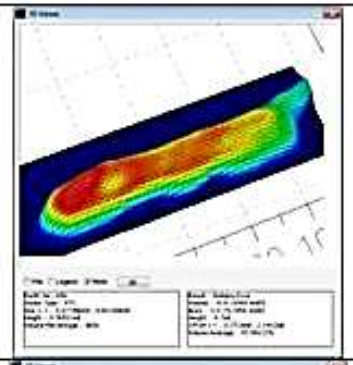
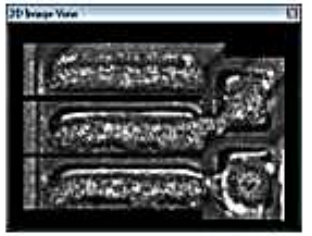
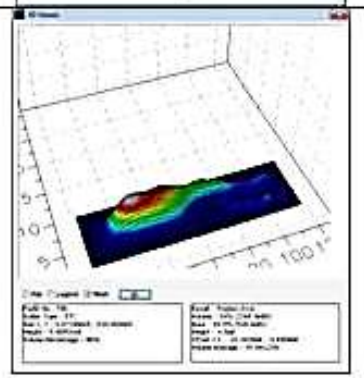
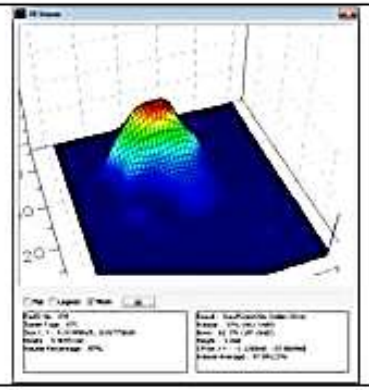




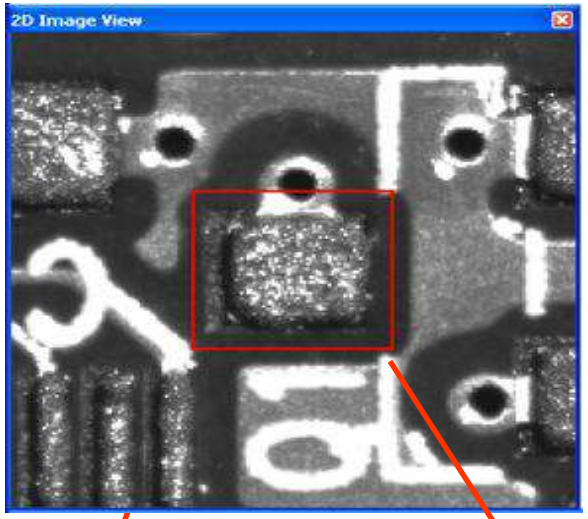
Position error

Defect Detection – Examples

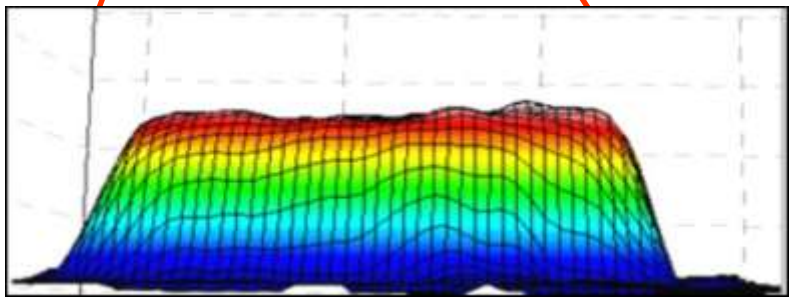
Defect Type	3D Images	2D Images
Insufficient Error		
Excessive Error		
Shape Error		

Defect Type	3D Images	2D Images
Bridging error		
Position Error		
Insufficient Error		

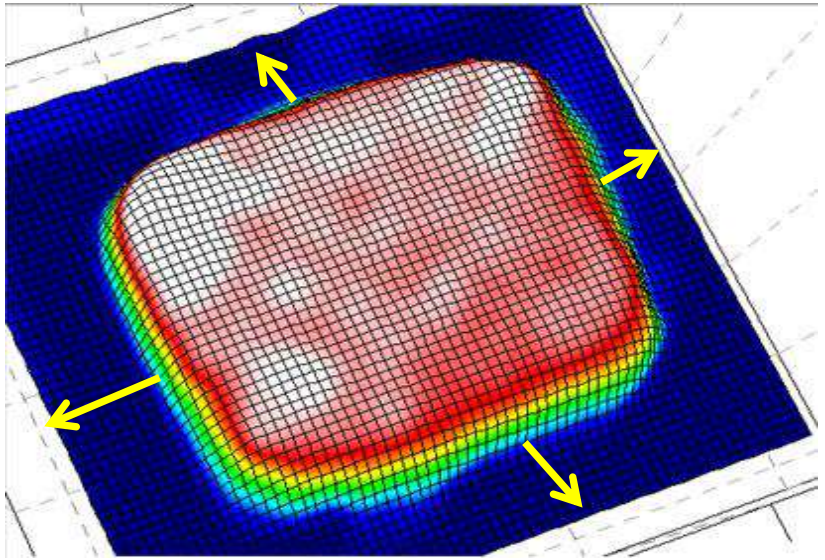
Measure Height per Pixel : Example for understanding purpose



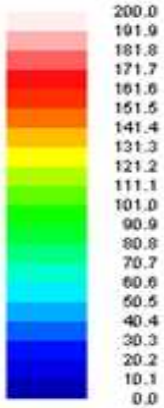
Extend 8 pixels



R O I



Legend:
Pixel's height (um)



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	10	10	10	10	10	10	10	10	10	10	10	10	10	10	20	20	20	20	20	20
2	10	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	10
3	10	40	40	40	40	50	50	50	50	50	60	60	60	60	60	40	40	40	40	10
4	10	70	110	110	120	120	120	120	120	120	120	120	130	130	130	130	130	130	130	10
5	10	70	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	80	10
6	10	70	140	140	150	160	160	160	180	180	160	160	160	160	160	160	160	150	80	10
7	10	70	140	140	150	160	160	160	190	200	180	160	160	160	160	160	160	150	90	10
8	10	70	150	160	160	160	160	160	170	170	160	160	160	160	160	160	160	150	90	10
9	10	70	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	90	10
10	10	70	140	140	140	140	140	140	140	140	140	140	130	130	130	130	130	130	100	10
11	10	70	130	130	130	130	130	130	130	130	130	130	130	130	130	130	130	130	100	10
12	10	40	40	40	40	50	50	50	50	50	60	60	60	60	60	40	40	40	40	10
13	10	30	30	30	30	40	40	40	40	40	30	30	30	30	30	30	30	30	30	10
14	10	10	10	10	10	20	20	20	20	20	10	10	10	10	10	10	10	10	10	10

Thank You All
For Your Kind Attention

Adibhatla Krishna Rao
NMTronics India Pvt.Ltd.